

SHALL appear, are provided below.

Table 107. OTA Software Image File Layout

Name	Type
FileIdentifier	uint32
TotalSize	uint64
HeaderSize	uint32
Header	see Appendix A, Tag-length-value (TLV) Encoding Format
Payload	N/A

11.21.2.1. FileIdentifier field

The FileIdentifier field is a fixed-width, little-endian-encoded, unsigned 32-bit value that SHALL be included at the beginning of all OTA software image files in order to quickly identify and distinguish the file as being of that format, without having to examine the contents of the whole file. This helps distinguishing the file from other file types in storage. The fixed constant value is defined to be 0x1BEEF11E.

11.21.2.2. TotalSize field

The TotalSize field is a fixed-width, little-endian-encoded, unsigned 64-bit value that SHALL indicate the total size, in bytes, of the entire file, including all fields and Payload. This field SHALL match the total stored size of the file. It SHALL match the sum of:

- The sizes of the FileIdentifier, TotalSize, and HeaderSize fields.
- The value of the HeaderSize field of this header.
- The value of the PayloadSize field of the Header subfield.

For example, given:

- Size of total Header structure is 105 bytes, reflected as HeaderSize = 105
- Payload size is 128KiB = 128 * 1024 bytes = 131072 bytes, reflected as Header.PayloadSize = 131072

Then the TotalSize would be the sum of:

- the size of the FileIdentifier, TotalSize, and HeaderSize fields: 4 + 8 + 4 = 16 bytes
- the HeaderSize value = 105 bytes
- the Header.PayloadSize value = 131072 bytes

This would give a total of 16 + 105 + 131072 = 131193 bytes. The overall file SHALL have this size and no more, and the TotalSize field SHALL contain that value.